

Distributional Semantic Models

Part 1: Introduction

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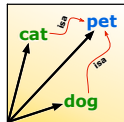
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<http://wordspace.collocations.de/doku.php/course:start>

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Outline

Introduction

- The distributional hypothesis
- Distributional semantic models
- DSMs and semantic similarity
- Three famous examples

Getting practical

- Software and further information
- R as a (toy) laboratory

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Introduction

The distributional hypothesis

Distributional semantic models

DSMs and semantic similarity

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Meaning & distribution

- ▶ “Die Bedeutung eines Wortes liegt in seinem Gebrauch.”
— Ludwig Wittgenstein
- ▶ “You shall know a word by the company it keeps!”
— J. R. Firth (1957)
- ▶ Distributional hypothesis: difference of meaning correlates with difference of distribution (Zellig Harris 1954)
- ▶ “What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it [...] in everyday discourse.” (Miller 1986)

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👉 semantic distance
- ▶ “What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it [...] in everyday discourse.” (Miller 1986)

What is the meaning of the mystery word?

Can we infer meaning from usage?

- ▶ He handed her her glass of [REDACTED].
- ▶ Beef dishes are made to complement the [REDACTED]s.
- ▶ Nigel staggered to his feet, face flushed from too much [REDACTED].
- ▶ Malbec, one of the lesser-known [REDACTED] grapes, responds well to Australia's sunshine.
- ▶ I dined off bread and cheese and this excellent [REDACTED].
- ▶ The drinks were delicious: blood-red [REDACTED] as well as light, sweet Rhenish.

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- 👉 [REDACTED] is a heavy red alcoholic beverage made from grapes

What is the meaning of the mystery word?

Can we infer meaning from usage?

- ▶ He handed her her glass of **claret** .
- ▶ Beef dishes are made to complement the **claret** s.
- ▶ Nigel staggered to his feet, face flushed from too much **claret** .
- ▶ Malbec, one of the lesser-known **claret** grapes, responds well to Australia's sunshine.
- ▶ I dined off bread and cheese and this excellent **claret** .
- ▶ The drinks were delicious: blood-red **claret** as well as light, sweet Rhenish.

 **claret** is a heavy red alcoholic beverage made from grapes

All examples from British National Corpus (handpicked and slightly edited).

Word sketch of “cat”

Can we infer meaning from collocations (as Firth suggests)?

cat British National Corpus freq = 5381

<https://sketchengine.eu/>

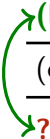
object of 964 2.0	and/or 1056 1.7	pp obj like-p 106 28.9	possessor 91 1.9	possession 232 4.7
skin 9 7.91	dog 208 8.49	grin 11 7.63	Schrödinger 8 10.87	cradle 24 9.91
diddle 7 7.85	cat 68 8.01	fight 9 4.62	witch 4 6.82	whisker 9 8.92
stroke 10 7.09	kitten 13 8.01	smile 4 4.24	gardener 4 6.0	paw 5 7.44
torture 5 6.57	fiddle 9 7.71	look 11 2.04	Henry 8 4.91	fur 9 7.14
feed 22 6.34	mouse 29 7.68		neighbour 5 4.28	tray 4 5.34
rain 4 6.3	monkey 15 7.55	pp among-p 17 14.8		tail 5 4.91
chase 9 6.27	budgie 4 6.74	pigeon 15 8.66		tongue 5 4.89
rescue 7 6.15	rabbit 12 6.48			ear 5 4.0

subject of 842 3.3	adj subject of 142 2.6	pp obj of-p 324 1.3	modifier 1622 1.2	modifies 610 0.5
purr 7 7.76	asleep 4 6.09	moral 4 7.06	pussy 76 10.42	flap 16 8.39
miaow 5 7.57	alive 4 5.06	breed 6 5.77	Cheshire 45 8.9	litter 15 8.15
mew 4 7.18	concerned 4 2.94	signal 4 3.89	stray 25 8.7	phobia 5 7.64
jump 20 6.95	black 4 2.36	sight 4 3.77	siamese 17 8.35	burglar 8 7.55
scratch 8 6.84	likely 4 1.96	species 5 3.36	tabby 17 8.35	faeces 6 7.47
leap 10 6.78		game 9 3.14	wild 53 7.94	assay 10 7.38
stalk 4 6.56		picture 6 2.99	pet 31 7.92	Hastings 7 6.91
react 4 5.33		death 7 2.71	tom 12 7.8	scan 4 6.59

A thought experiment: deciphering hieroglyphs

							
(knife)		51	20	84	0	3	0
(cat)		52	58	4	4	6	26
???		115	83	10	42	33	17
(boat)		59	39	23	4	0	0
(cup)		98	14	6	2	1	0
(pig)		12	17	3	2	9	27
(banana)		11	2	2	0	18	0

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$$\text{sim}(\text{triangle, vertical bar, triangle}, \text{wavy line, triangle, vertical bar, triangle}) = 0.770$$

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$$\text{sim}(\text{hieroglyph: unknown}, \text{hieroglyph: pig}) = 0.939$$

A thought experiment: deciphering hieroglyphs

							
(knife)		51	20	84	0	3	0
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(cup)		98	14	6	2	1	0
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(banana)		11	2	2	0	18	0

$$\text{sim}(\text{eye in box}, \text{cat}) = 0.961$$

English as seen by the computer ...

		get 	see 	use 	hear 	eat 	kill 
knife		51	20	84	0	3	0
cat		52	58	4	4	6	26
dog		115	83	10	42	33	17
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verb-object counts from British National Corpus

Geometric interpretation

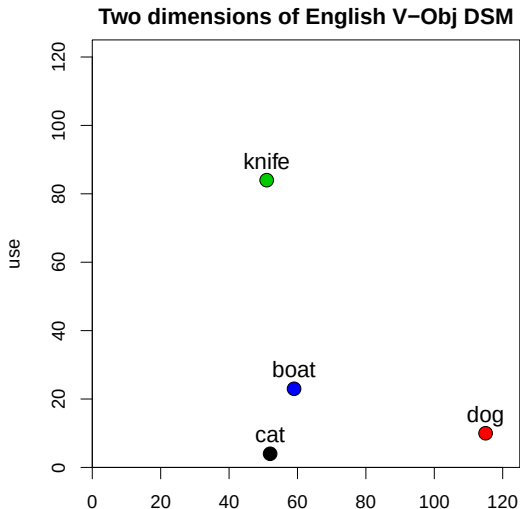
- ▶ row vector $\mathbf{x}_{\text{dog}}$ describes usage of word *dog* in the corpus
- ▶ can be seen as coordinates of point in n -dimensional Euclidean space

	get	see	use	hear	eat	kill
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co-occurrence matrix M

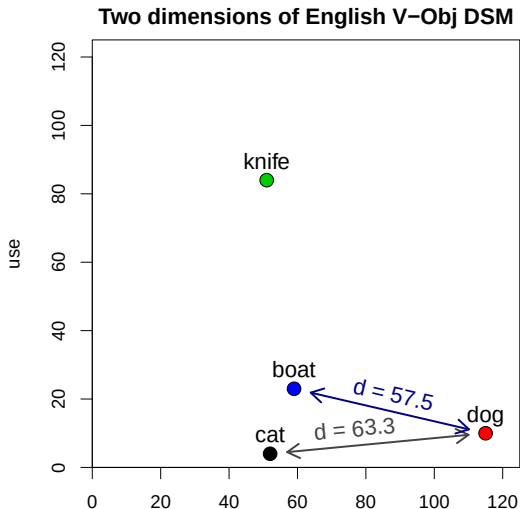
Geometric interpretation

- ▶ row vector $\mathbf{x}_{\text{dog}}$ describes usage of word *dog* in the corpus
- ▶ can be seen as coordinates of point in n -dimensional Euclidean space
- ▶ illustrated for two dimensions: *get* and *use*
- ▶ $\mathbf{x}_{\text{dog}} = (115, 10)$



Geometric interpretation

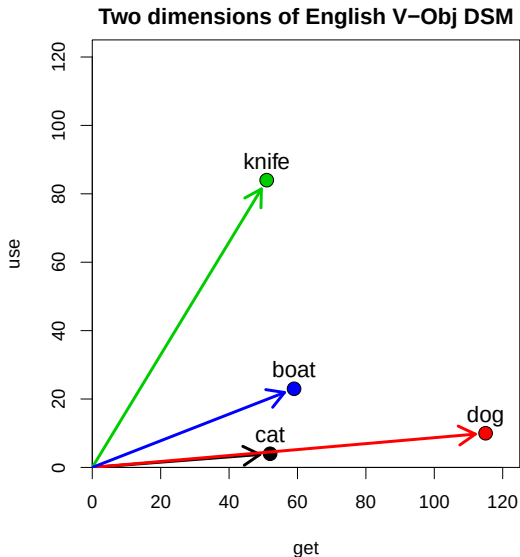
- ▶ similarity = spatial proximity (Euclidean dist.)
- ▶ location depends on frequency of noun ($f_{\text{dog}} \approx 2.7 \cdot f_{\text{cat}}$)



get

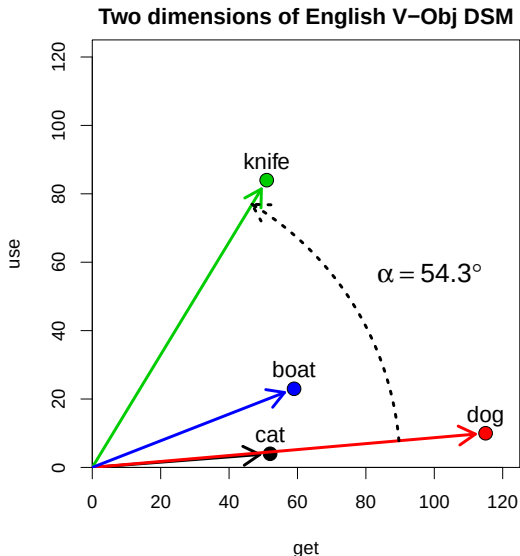
Geometric interpretation

- ▶ vector can also be understood as arrow from origin
- ▶ direction more important than location



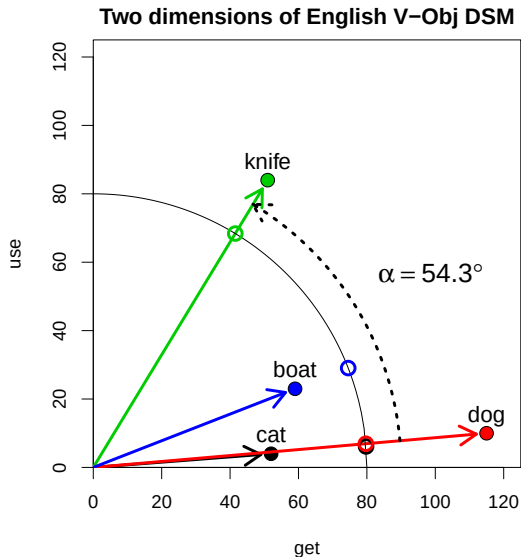
Geometric interpretation

- ▶ vector can also be understood as arrow from origin
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- ▶ use angle α as distance measure



Geometric interpretation

- ▶ vector can also be understood as arrow from origin
- ▶ direction more important than location
- ▶ use angle α as distance measure
- ▶ or normalise length $\|\mathbf{x}_{\text{dog}}\|$ of arrow



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DSMs and semantic similarity

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General definition of DSMs

A **distributional semantic model** (DSM) is a scaled and/or transformed co-occurrence matrix $\mathbf{M}$, such that each row $\mathbf{x}$ represents the distribution of a target term across contexts.

	get	see	use	hear	eat	kill
knife	0.027	-0.024	0.206	-0.022	-0.044	-0.042
cat	0.031	0.143	-0.243	-0.015	-0.009	0.131
dog	-0.026	0.021	-0.212	0.064	0.013	0.014
boat	-0.022	0.009	-0.044	-0.040	-0.074	-0.042
cup	-0.014	-0.173	-0.249	-0.099	-0.119	-0.042
pig	-0.069	0.094	-0.158	0.000	0.094	0.265
banana	0.047	-0.139	-0.104	-0.022	0.267	-0.042

Term = word, lemma, phrase, morpheme, word pair, ...

Nearest neighbours

DSM based on verb-object relations from BNC, reduced to 100 dim. with SVD

Neighbours of **trousers** (cosine angle):

☞ shirt (18.5), blouse (21.9), scarf (23.4), jeans (24.7), skirt (25.9), sock (26.2), shorts (26.3), jacket (27.8), glove (28.1), coat (28.8), cloak (28.9), hat (29.1), tunic (29.3), overcoat (29.4), pants (29.8), helmet (30.4), apron (30.5), robe (30.6), mask (30.8), tracksuit (31.0), jersey (31.6), shawl (31.6), ...

Nearest neighbours

DSM based on verb-object relations from BNC, reduced to 100 dim. with SVD

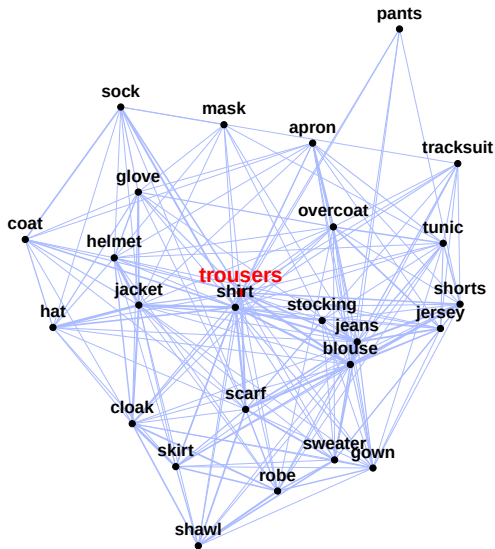
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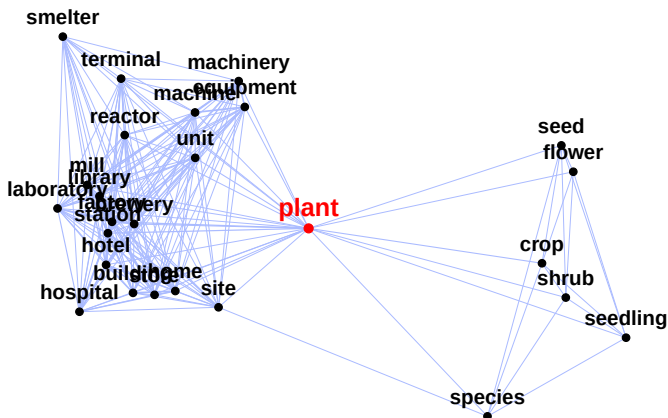
Neighbours of **rage** (cosine angle):

👉 anger (28.5), fury (32.5), sadness (37.0), disgust (37.4), emotion (39.0), jealousy (40.0), grief (40.4), irritation (40.7), revulsion (40.7), scorn (40.7), panic (40.8), bitterness (41.6), resentment (41.8), indignation (41.9), excitement (42.0), hatred (42.5), envy (42.8), disappointment (42.9), ...

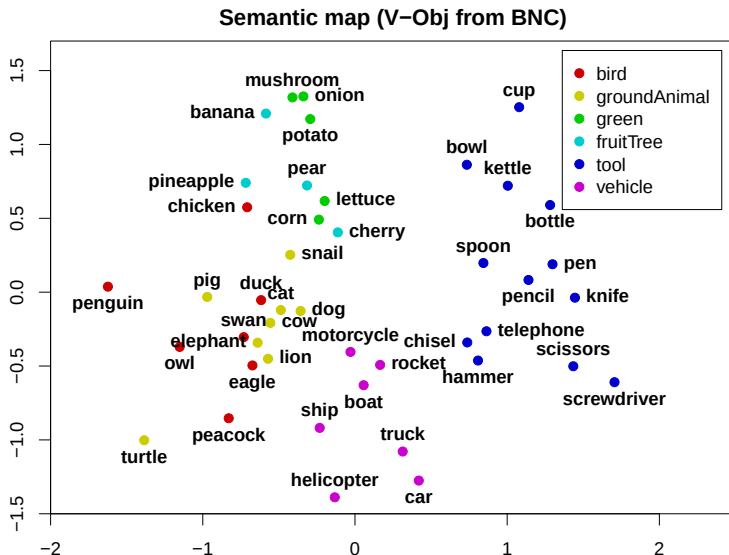
Nearest neighbours with similarity graph



Nearest neighbours with similarity graph

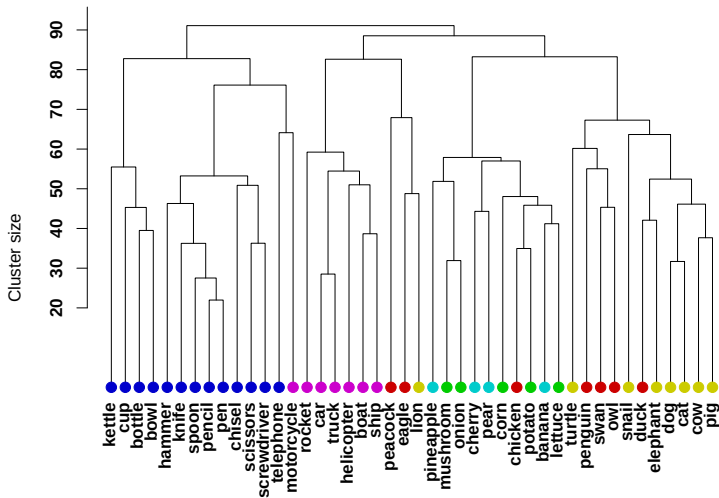


Semantic maps



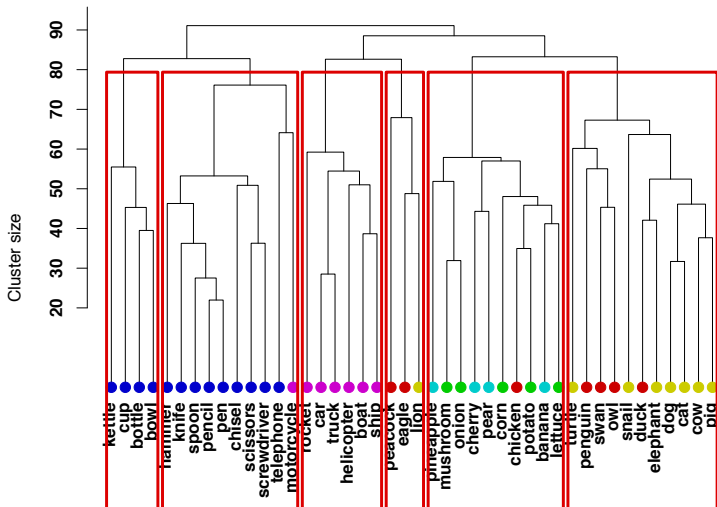
Clustering

Clustering of concrete nouns (V-Obj from BNC)

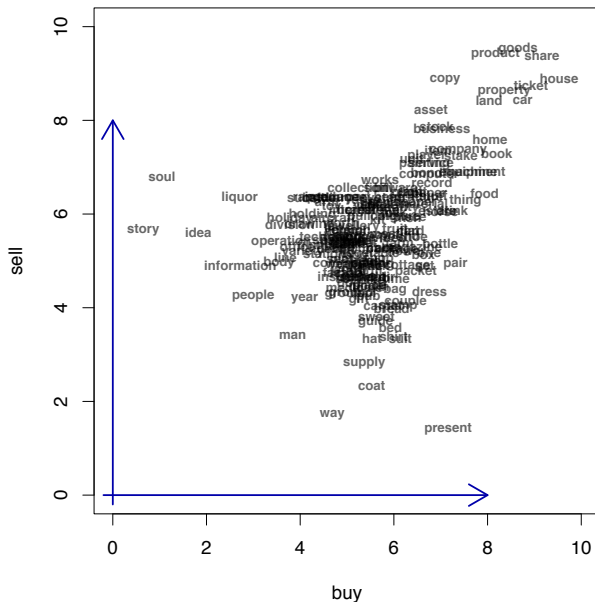


Clustering

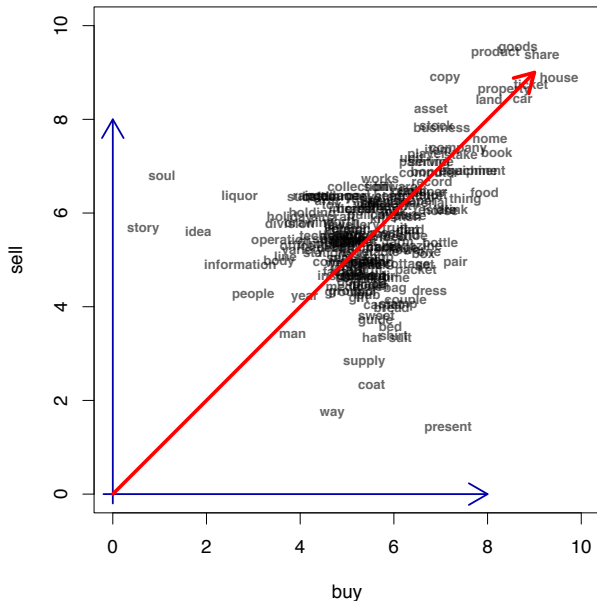
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Latent “meaning” dimensions



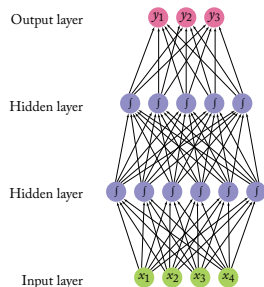
Latent “meaning” dimensions



DSM vectors as word embeddings

DSM vector as sub-symbolic meaning representation

- ▶ feature vector for machine learning algorithm
 - ▶ input for neural network
 - ▶ such **distributed** representations are known as **embeddings**
- 👉 embeddings $\nRightarrow$ distributional



(Goldberg 2017, Fig. 4.2)

DSM vectors as word embeddings

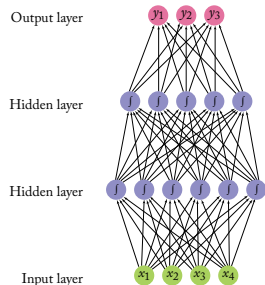
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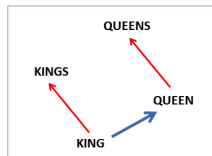
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Computation in semantic space

- ▶ find meaningful subdimensions in DSM space (→ correlation)
- ▶ linear operations on vectors



(Goldberg 2017, Fig. 4.2)

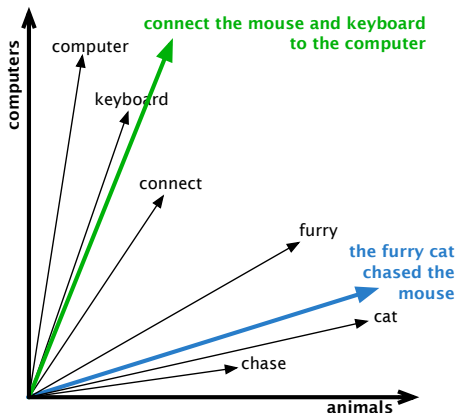


(Mikolov et al. 2013c, Fig. 2)

DSM vectors as context representation

Context vectors for word tokens
(Schütze 1998)

- ▶ **bag-of-words** approach:
centroid of all context words in
the sentence
- ▶ application to WSD



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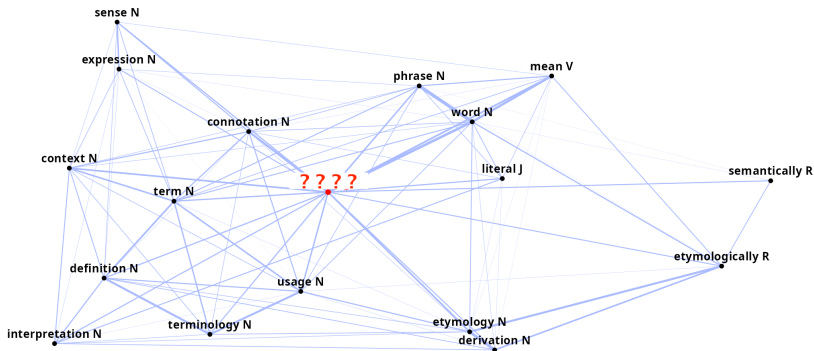
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Inverse distributional semantics

Which word occurs in similar contexts as the ones displayed in this graph?



... look at the neighbours:
is there one notion of similarity that explains all of them?

Distributional similarity as semantic similarity

- ▶ DSM similarity as a **quantitative notion** (Harris 1954)
 - ▶ if **a** is closer to **b** than to **c** in the distributional vector space, then *a* is more semantically similar to *b* than to *c*
- ▶ DSM similarity as a **graded notion**, differently from **categorical** nature of most theoretical accounts
- ▶ DSM similarity as the empirical correlate of a **heterogeneous set of phenomena**

... which we may want to tease apart!
- ▶ DSM similarity is **symmetric** – cognition is not

... can we fix this?

Characterising DSM similarity

- ▶ DSMs are often thought to represent **taxonomic** similarity
 - ▶ words that tend to occur in the same contexts
- ▶ Words that share many contexts share many properties (attributes) and are thus **taxonomically/ontologically similar**
 - ▶ synonyms (*rhino/rhinoceros*)
 - ▶ antonyms and values on a scale (*good/bad*)
 - ▶ co-hyponyms (*rock/jazz*)
 - ▶ hyper- and hyponyms (*rock/basalt*)
- ▶ Taxonomic similarity is seen as the **fundamental semantic relation** organising the vocabulary of a language, allowing categorisation, generalisation and inheritance...

Is distributional similarity *just* taxonomic?

Nearest DSM neighbors have different types of **semantic relations**.

car (BNC, L2/R2 span)

- ▶ van **co-hyponym**
- ▶ vehicle **hyperonym**
- ▶ truck **co-hyponym**
- ▶ motorcycle **co-hyponym**
- ▶ driver **related entity**
- ▶ motor **part**
- ▶ lorry **co-hyponym**
- ▶ motorist **related entity**
- ▶ cavalier **hyponym**
- ▶ bike **co-hyponym**

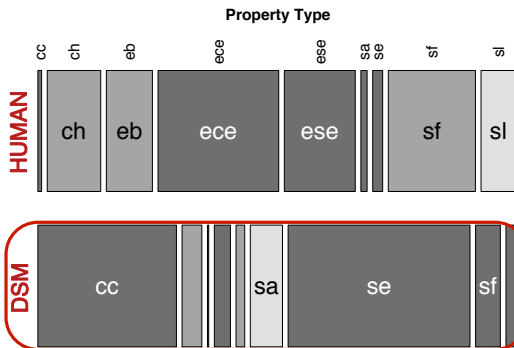
car (BNC, L30/R30 span)

- ▶ drive **function**
- ▶ park **typical action**
- ▶ bonnet **part**
- ▶ windscreen **part**
- ▶ hatchback **part**
- ▶ headlight **part**
- ▶ jaguar **hyponym**
- ▶ garage **location**
- ▶ cavalier **hyponym**
- ▶ tyre **part**

<http://clic.cimec.unitn.it/infomap-query/>

Is distributional similarity *just* taxonomic?

Manual annotation: what are the properties of *car*? Humans vs DSM



Taxonomic category:

cc (co-)hyponym *truck*

ch hyponym *vehicle*

Properties of entity:

eb typical behaviour

ece ext. component *wheel*

ese surf. property *smooth*

Situationally associated:

sa action *park*

se other entity *traffic light*

sf function *drive*

sl location *garage*

sp participant *driver*

Task: humans: given a word, generate properties; DSM: generate top 10 neighbors. Items: 44 concrete English nouns (Baroni & Lenci 2008).

DSM similarity: concepts & terminology

Attributional similarity vs. semantic relatedness

- ▶ **Attributional similarity** (← taxonomic) – two words sharing a large number of salient features (attributes)
 - ▶ **synonymy** (*car/automobile*)
 - ▶ **co-hyponymy** (*car/van/truck*)
 - ▶ **hyperonymy** (*car/vehicle*)
 - 👉 subset → distributional inclusion (Lenci & Benotto 2012)
 - ▶ **antonymy** (*hot/cold*)
 - 👉 intuitively they are the opposite of similar, and yet ...

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- ▶ **Semantic relatedness** (Budanitsky & Hirst 2006) – two words semantically associated without necessarily being similar
 - ▶ function (*car/drive*)
 - ▶ meronymy (*car/tyre*)
 - ▶ location (*car/road*)
 - ▶ attribute (*car/fast*)

☞ Why similar in DSMs?

DSM similarity: concepts & terminology

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☞ Why similar in DSMs? co-occur → overlapping context spans

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Attributional vs. relational similarity

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- ▶ **Relational similarity** (Turney 2006) – similar relation between pairs of words (analogy)
 - ▶ *policeman:gun :: teacher:book*
 - ▶ *mason:stone :: carpenter:wood*
 - ▶ *traffic:street :: water:riverbed*

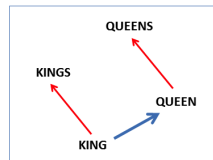
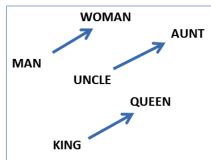
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Textbook example of neural embeddings application



Problem: symmetry in DSM similarity

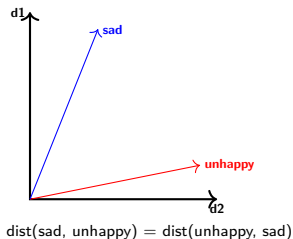
Symmetric similarity does not fit all phenomena

- ▶ Motivation: cognitive processes are notoriously asymmetric (and so is human intuition about semantic similarity)
- ▶ Solution: use asymmetric measure of distributional similarity

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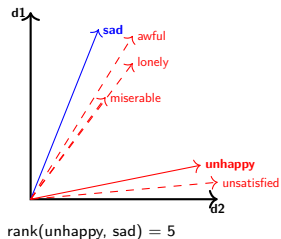
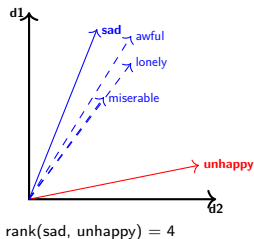
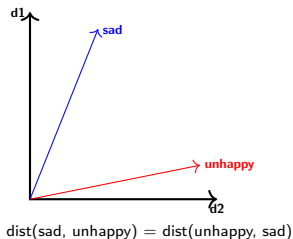
- ▶ Motivation: cognitive processes are notoriously asymmetric (and so is human intuition about semantic similarity)
- ▶ Solution: use asymmetric measure of distributional similarity
- ▶ Solution: use neighbour rank instead of similarity score
 - ▶ makes predictions comparable across models and measures
 - ▶ intuition: different density of semantic neighbourhood



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DSMs and semantic similarity

Three famous examples

Getting practical

Software and further information

R as a (toy) laboratory

Latent Semantic Analysis (Landauer & Dumais 1997)

- ▶ Corpus: 30,473 articles from Grolier's *Academic American Encyclopedia* (4.6 million words in total)
 - 👉 articles were limited to first 2,000 characters
- ▶ Word-article frequency matrix for 60,768 words
 - ▶ row vector shows frequency of word in each article
- ▶ Logarithmic frequencies scaled by word entropy
- ▶ Reduced to 300 dim. by singular value decomposition (SVD)
 - ▶ borrowed from LSI (Dumais *et al.* 1988)
 - 👉 central claim: SVD reveals latent semantic features, not just a data reduction technique
- ▶ Evaluated on TOEFL synonym test (80 items)
 - ▶ LSA model achieved 64.4% correct answers
 - ▶ also simulation of learning rate based on TOEFL results

Word Space (Schütze 1992, 1993, 1998)

- ▶ Corpus: $\approx$ 60 million words of news messages
 - ▶ from the *New York Times* News Service
- ▶ Word-word co-occurrence matrix
 - ▶ 20,000 target words & 2,000 context words as features
 - ▶ row vector records how often each context word occurs close to the target word (co-occurrence)
 - ▶ co-occurrence window: left/right 50 words (Schütze 1998) or $\approx$ 1000 characters (Schütze 1992)
- ▶ Rows weighted by inverse document frequency (tf.idf)
- ▶ Context vector = centroid of word vectors (bag-of-words)
 - 👉 goal: determine “meaning” of a context
- ▶ Reduced to 100 SVD dimensions (mainly for efficiency)
- ▶ Evaluated on unsupervised word sense induction by clustering of context vectors (for an ambiguous word)
 - ▶ induced word senses improve information retrieval performance

HAL (Lund & Burgess 1996)

- ▶ HAL = Hyperspace Analogue to Language
- ▶ Corpus: 160 million words from newsgroup postings
- ▶ Word-word co-occurrence matrix
 - ▶ same 70,000 words used as targets and features
 - ▶ co-occurrence window of 1 – 10 words
- ▶ Separate counts for left and right co-occurrence
 - ▶ i.e. the context is *structured*
- ▶ In later work, co-occurrences are weighted by (inverse) distance (Li *et al.* 2000)
 - ▶ but no dimensionality reduction
- ▶ Applications include construction of semantic vocabulary maps by multidimensional scaling to 2 dimensions

HAL (Lund & Burgess 1996)

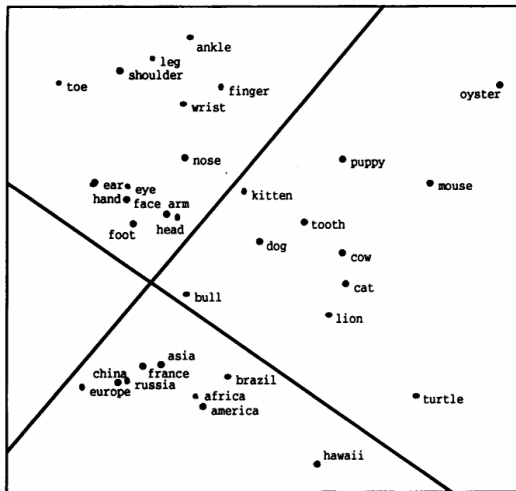


Figure 2. Multidimensional scaling of co-occurrence vectors.

word2vec embeddings (Mikolov *et al.* 2013a,b)

- ▶ Embeddings are learned by a (simple) neural model rather than obtained directly as distributional vectors
 - ▶ known as **predict** rather than **count** models (Baroni *et al.* 2014)
- ▶ Corpus: Google News, 6 billion words
- ▶ Learning objective of the skip-gram model (SGNS)
 - ▶ predict conditional probability distribution $\Pr(c \mid t)$ for collocational profile of target word t
 - ▶ based on learned target embeddings $\mathbf{w}_t$ and (ephemeral) collocate embeddings $\mathbf{v}_c$
 - 👉 can be understood as factorisation of co-occurrence matrix (Levy & Goldberg 2014; Pennington *et al.* 2014)
- ▶ Evaluation focuses on syntactic and semantic analogy tasks

Many parameters . . .

- ▶ Enormous range of DSM parameters and applications
- ▶ Examples showed four entirely different models, each tuned to its particular application

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 - ▶ part 3: Evaluating DSM representations
 - ▶ part 4: Understanding distributional semantics
 - ▶ part 5: Mathematical foundations (matrix algebra)

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 - ▶ part 5: Mathematical foundations (matrix algebra)
- ➡ Distributional semantics is an empirical science

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Some applications in computational linguistics

- ▶ Query expansion in information retrieval (Grefenstette 1994)
- ▶ Unsupervised part-of-speech induction (Schütze 1995)
- ▶ Word sense disambiguation (Schütze 1998; Rapp 2004b)
- ▶ Thesaurus compilation (Lin 1998; Rapp 2004a)
- ▶ Attachment disambiguation (Pantel & Lin 2000)
- ▶ Probabilistic language models (Bengio *et al.* 2003)
- ▶ Translation equivalents (Sahlgren & Karlgren 2005)
- ▶ Ontology & wordnet expansion (Pantel *et al.* 2009)
- ▶ Language change (Sagi *et al.* 2009; Hamilton *et al.* 2016)
- ▶ Multiword expressions (Kielia & Clark 2013)
- ▶ Analogies (Turney 2013; Gladkova *et al.* 2016)
- ▶ Sentiment analysis (Rothe & Schütze 2016; Yu *et al.* 2017)
- 🔗 Input representation for neural networks & machine learning

Software packages

Infomap NLP	C	<i>classical LSA-style DSM</i>
HiDEx	C++	<i>re-implementation of the HAL model (Lund & Burgess 1996)</i>
SemanticVectors	Java	<i>scalable architecture based on random indexing representation</i>
S-Space	Java	<i>complex object-oriented framework</i>
JoBimText	Java	<i>UIMA / Hadoop framework</i>
Gensim	Python	<i>complex framework, focus on parallelization and out-of-core algorithms</i>
Vecto	Python	<i>framework for count & predict models</i>
DISSECT	Python	<i>user-friendly, designed for research on compositional semantics</i>
wordspace	R	<i>interactive research laboratory, but scales to real-life data sets</i>
text2vec	R	<i>GloVe embeddings & topic models</i>

click on package name to open Web page

Further information

- ▶ Handouts & other materials available from wordspace wiki at <http://wordspace.collocations.de/>
- ▶ Tutorial is open source (CC) and can be downloaded from <http://r-forge.r-project.org/projects/wordspace/>

Further information

- ▶ Review papers on distributional semantics
 - ▶ Turney, Peter D. and Pantel, Patrick (2010). *From frequency to meaning: Vector space models of semantics*. *Journal of Artificial Intelligence Research*, 37, 141–188.
 - ▶ Erk, Katrin (2012). *Vector Space Models of Word Meaning and Phrase Meaning: A Survey*. *Language and Linguistics Compass*, 6-1, 635–653.
 - ▶ Boleda, Gemma (2020). *Distributional Semantics and Linguistic Theory*. *Annual Review of Linguistics*, 6-1, 213–234.
- ▶ Textbook on distributional semantics
 - ▶ Lenci, Alessandro and Sahlgren, Magnus (2023). *Distributional Semantics*. Studies in Natural Language Processing. Cambridge University Press, Cambridge.

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Prepare to get your hands dirty ...

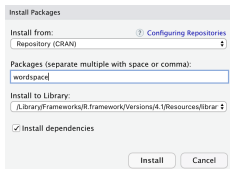
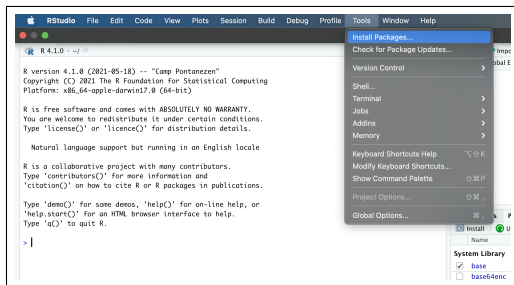
- ▶ We will use the statistical programming environment **R** as a toy laboratory in this tutorial
 - 👉 but one that scales to real-life applications

Software installation

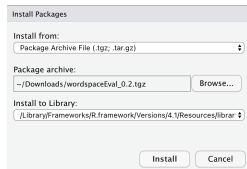
- ▶ **R** version 4.0 or newer from <http://www.r-project.org/>
- ▶ **RStudio** from <http://www.rstudio.com/>
- ▶ R packages from CRAN (through RStudio menu)
 - ▶ `sparsesvd`, `wordspace`
 - ▶ recommended: `e1071`, `rsparse`, `Rtsne`, `uwot`
 - ▶ optional: `tm`, `quanteda`, `data.table`, `wordcloud`, `shiny`, `spacyr`, `udpipe`, `coreNLP`
- ▶ Get data sets, precompiled DSMs and `wordspaceEval` package (with some non-public data sets) from <http://wordspace.collocations.de/doku.php/course:material>

Prepare to get your hands dirty . . .

Installing workspace and workspaceEval in RStudio



workspace



workspaceEval

Prepare to get your hands dirty . . .

Setting up a working directory and RStudio project

- ▶ Create a separate **directory** (folder) for this course
 - ▶ subdirectory **models** for pre-compiled DSMs (large files)
 - ▶ subdirectory **data** for other data files
- ▶ Recommended: set up **RStudio project** for the course
 - ▶ click *New Project* (top right corner), then *Existing Directory*
 - ▶ choose the course directory you've just created
 - ▶ this will be set as your R working directory within the project!
 - 👉 you can easily switch between different RStudio projects
- ▶ Alternatively: set working directory at start of session
 - ▶ e.g. `setwd("~/zellig/Desktop/dsm_tutorial")`
- ▶ Work with **R scripts** rather than in interactive console
 - ▶ RStudio: add *R Script* from drop-down menu in top left corner
 - ▶ we provide example scripts for each part of the tutorial

First steps in R

Start each session by loading the workspace package.

```
> library(workspace)
```

The package includes various example data sets, some of which should look familiar to you.

```
> DSM_HieroglyphsMatrix
```

	get	see	use	hear	eat	kill
knife	51	20	84	0	3	0
cat	52	58	4	4	6	26
dog	115	83	10	42	33	17
boat	59	39	23	4	0	0
cup	98	14	6	2	1	0
pig	12	17	3	2	9	27
banana	11	2	2	0	18	0

Term-term matrix

Term-term matrix records co-occurrence frequencies with feature terms for each target term

```
> DSM_TermTermMatrix
```

	<i>breed</i>	<i>tail</i>	<i>feed</i>	<i>kill</i>	<i>important</i>	<i>explain</i>	<i>likely</i>
cat	83	17	7	37	–	1	–
dog	561	13	30	60	1	2	4
animal	42	10	109	134	13	5	5
time	19	9	29	117	81	34	109
reason	1	–	2	14	68	140	47
cause	–	1	–	4	55	34	55
effect	–	–	1	6	60	35	17

Term-context matrix

Term-context matrix records frequency of term in each individual context (e.g. sentence, document, Web page, encyclopaedia article)

```
> DSM_TermContextMatrix
```

	<i>Felidae</i>	<i>Pet</i>	<i>Feral</i>	<i>Bloat</i>	<i>Philosophy</i>	<i>Kant</i>	<i>Back pain</i>
cat	10	10	7	–	–	–	–
dog	–	10	4	11	–	–	–
animal	2	15	10	2	–	–	–
time	1	–	–	–	2	1	–
reason	–	1	–	–	1	4	1
cause	–	–	–	2	1	2	6
effect	–	–	–	1	–	1	–

Some basic operations on a DSM matrix

```
# apply log-transformation to de-skew co-occurrence frequencies
> M <- log2(DSM_HieroglyphsMatrix + 1) # see part 2
> round(M, 3)

# compute distributional similarity (cosine measure)
> pair.distances("dog", "cat", M, convert=FALSE)
  dog/cat
0.9610952

# find nearest neighbours (defaults to angles)
> nearest.neighbours(M, "dog", n=3)
      cat      pig      cup
16.03458 20.08826 31.77784

> plot(nearest.neighbours(M, "dog", n=3, dist.matrix=TRUE))
```

Playing with a larger model

Term-term matrix, dimensionality-reduced, built from Web texts for target words in the format *lemma_POS* (e.g. banana_N)

```
> DSM_Vectors  
> View(DSM_Vectors)
```

Let's inspect some nearest neighbors:

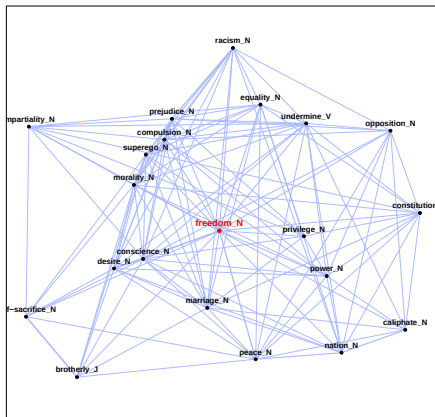
```
> nearest.neighbours(DSM_Vectors, "banana_N", n=4)  
coconut_N  pineapple_N  watermelon_N      bean_N  
10.86118    12.60826     13.35160     13.79671
```

```
> nearest.neighbours(DSM_Vectors, "freedom_N", n=4)  
peace_N    morality_N    equality_N  conscience_N  
30.13420   34.18397    34.23418   34.23894
```

Playing with a larger model

Or create a semantic map for a word we are interested in:

```
> plot(nearest.neighbours(DSM_Vectors, "freedom_N", n=20,  
  dist.matrix=TRUE))
```



... and with an even larger model

You can download several **large pre-compiled DSMs** from the course wiki, which represent different parameters of the co-occurrence matrix (→ part 2).

- ▶ e.g. WP500_DepFilter_Lemma.rda
- ▶ download this file to subdirectory models

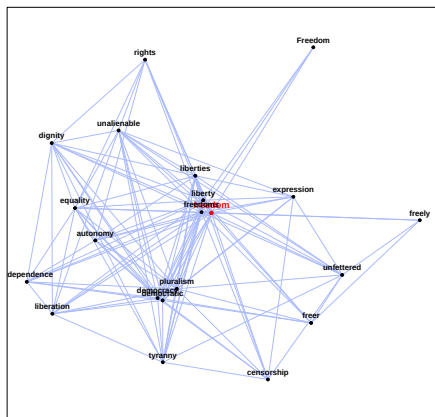
```
> load("models/WP500_DepFilter_Lemma.rda", verbose=TRUE)
Loading objects:
  WP500_DepFilter_Lemma
> model <- WP500_DepFilter_Lemma # assign to a shorter name
```

Now try the semantic map again:

```
> plot(nearest.neighbours(model, "freedom_N", n=20,
                           dist.matrix=TRUE))
```

Freedom in a neural embedding model: word2vec

```
> load("GoogleNews300_wf200k.rda", verbose=TRUE)
> embeddings <- GoogleNews300_wf200k.rda
> plot(nearest.neighbours(embeddings, "freedom", n=20,
                           dist.matrix=TRUE))
```



Explorations

While you wait for Part 2 of the tutorial,
you can explore some DSM similarity networks online:

- ▶ <https://corpora.linguistik.uni-erlangen.de/shiny/workspace/>
- ▶ built in R with `workspace` and `shiny`

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